

# **PURCHASE PATTERN ANALYSIS USING MARKET BASKET ANALYSIS AND APRIORI ALGORITHM**

## **Project Report**

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# 1. EXECUTIVE SUMMARY

Purchase Pattern Analysis is a retail analytics project designed to understand customer purchasing behavior and identify relationships between products purchased together.

The project uses a retail transaction dataset containing **50,000 transaction records and 7 columns**. The analysis begins with exploratory data analysis and data cleaning to identify missing values, duplicate records, incorrect data types, and invalid quantities and prices.

After preprocessing, the dataset was reduced to **49,152 valid records**. A transaction-level basket was then created using the `BillNo` and `Itemname` fields. The basket was transformed into a binary format where each product is represented by whether it was purchased in a particular transaction.

The **Apriori algorithm** was applied to identify frequent itemsets. Association rules were then generated using **support, confidence, and lift**. The analysis used a minimum support of **0.02** and a minimum lift threshold of **1**.

The resulting rules helped identify products that are frequently purchased together and products that have strong predictive relationships. Some rules achieved confidence above 0.60 and very high lift values, indicating strong associations between particular products.

The findings can support business activities such as cross-selling, product bundling, targeted promotions, inventory planning, and product placement.

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# 2. INTRODUCTION

Retail businesses generate large volumes of transactional data every day. Each transaction contains valuable information about products purchased by customers, purchase quantities, prices, dates, and customer or geographical information.

Simply knowing which products sell the most does not fully explain customer purchasing behavior. Businesses also need to understand which products are frequently purchased together.

For example, if customers frequently purchase Product A together with Product B, the retailer can recommend Product B when a customer purchases Product A. This can increase cross-selling opportunities and improve the customer shopping experience.

This project applies **Market Basket Analysis (MBA)** and the **Apriori algorithm** to identify such purchasing relationships.

The project combines:

- Data cleaning

- Exploratory Data Analysis
- Transaction analysis
- Market Basket Analysis
- Apriori frequent-itemset mining
- Association-rule generation
- Power BI visualization
- Business recommendations

The overall objective is to convert raw transaction data into actionable business insights.

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### 3. PROJECT BACKGROUND

The project uses a retail transaction dataset containing records of customer purchases across products and countries.

The dataset contains information such as:

- Transaction/Bill number
- Product name
- Quantity purchased
- Transaction date
- Product price
- Customer ID
- Country

The original dataset contains **50,000 records**, covering transactions from **December 2010 to January 2011**. The United Kingdom represents the primary market in the dataset.

The analysis was performed using Python for data preparation and Market Basket Analysis, followed by visualization using Power BI.

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### 4. PROBLEM STATEMENT

Retailers often have large amounts of transaction data but may not know which products customers tend to purchase together.

Traditional sales analysis mainly focuses on individual product performance. However, understanding product associations can provide additional opportunities for:

- Cross-selling
- Product bundling

- Personalized recommendations
- Promotional campaigns
- Inventory management
- Store/product placement

Therefore, the problem addressed by this project is:

“How can retail transaction data be analyzed to identify frequent product combinations and meaningful product associations that can support sales and marketing decisions?”

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## 5. PROJECT OBJECTIVES

The main objectives of the project are:

1. To understand customer purchasing behavior using retail transaction data.
2. To perform Exploratory Data Analysis (EDA) on the transaction dataset.
3. To identify missing values, duplicates, invalid quantities, invalid prices, and other data-quality issues.
4. To clean and transform the raw dataset into an analysis-ready format.
5. To create transaction-level product baskets.
6. To apply the Apriori algorithm to identify frequent itemsets.
7. To generate association rules using support, confidence, and lift.
8. To identify strong relationships between products.
9. To visualize important analytical results using Power BI.
10. To derive actionable business recommendations for cross-selling, bundling, marketing, and inventory management.

These objectives are consistent with the project presentation and analytical notebook.

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## 6. SCOPE OF THE PROJECT

The project focuses on transaction-level retail purchase behavior.

**The project covers:**

- Retail transaction analysis
- Product frequency analysis

- Data quality analysis
- Data preprocessing
- Transaction basket creation
- Market Basket Analysis
- Apriori algorithm
- Association rules
- Support analysis
- Confidence analysis
- Lift analysis
- Product relationship analysis
- Power BI visualization
- Business recommendations

**The project does not cover:**

- Customer demographic segmentation
- Predictive sales forecasting
- Customer lifetime value prediction
- Price optimization
- Recommendation-system deployment
- Real-time transaction processing

These could be considered as extensions of the project in the future.

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## 7. DATASET DESCRIPTION

### 7.1 Dataset Overview

The project uses the retail transaction dataset **12 PPana.csv**.

#### Raw Dataset

- Records: **50,000**
- Columns: **7**
- Unique products: **2,802**
- Countries: **19**
- Date observations: **1,762 unique timestamps**

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### 7.2 Dataset Variables

| Column | Description |
|--------|-------------|
|--------|-------------|

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| Column       | Description                            |
|--------------|----------------------------------------|
| BillNo       | Unique transaction/bill number         |
| Itemname     | Name of the purchased product          |
| Quantity     | Number of units purchased              |
| Present_Date | Transaction date and time              |
| Price        | Unit price of the product              |
| CustomerID   | Customer identification number         |
| Country      | Country where the transaction occurred |

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## 8. TOOLS AND TECHNOLOGIES USED

### Python

Python was used as the primary programming language for data analysis.

### Pandas

Used for:

- Data loading
- Data manipulation
- Missing-value analysis
- Duplicate detection
- Data cleaning
- Aggregation
- Transformation

### NumPy

Used for numerical operations.

### Matplotlib

Used for data visualization.

### Seaborn

Used for statistical and graphical visualization.

The notebook imports Pandas, NumPy, Seaborn, and Matplotlib.

### Mlxtend

The `mlxtend.frequent_patterns` module was used for:

- Apriori algorithm
- Association-rule generation

## Power BI

Power BI was used to create visual representations of the analytical results, including product relationships and association-rule metrics.

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## 9. PROJECT METHODOLOGY

The project follows the following analytical workflow:

**Raw Dataset**

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**Data Understanding**

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**Exploratory Data Analysis**

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**Data Cleaning**

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**Data Transformation**

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**Transaction Basket Creation**

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**One-Hot/Binary Encoding**

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**Apriori Algorithm**

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**Frequent Itemsets**

↓

**Association Rules**

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**Support / Confidence / Lift Analysis**



**Power BI Visualization**



**Business Insights**



**Recommendations**

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## **10. DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS**

The first step was to understand the structure and quality of the raw dataset.

The notebook examined:

- Dataset shape
  - Data types
  - Missing values
  - Duplicate records
  - Unique values
  - Summary statistics
  - Product frequencies
  - Quantity distribution
  - Price distribution
- 

### **10.1 Dataset Shape**

The original dataset contained:

**50,000 rows × 7 columns**

The dataset contained 2,802 unique product names and 19 countries.

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### **10.2 Missing Values**

The raw dataset contained missing values in:



- Itemname: 145
- CustomerID: 18,568

The other columns did not contain missing values according to the notebook's initial missing-value analysis.

The EDA report summarizes the raw dataset as having **18,713 missing values in total**.

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## 10.3 Duplicate Records

The analysis identified:

**548 duplicate rows**

These duplicate records were removed during preprocessing.

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## 10.4 Product Analysis

The analysis identified the most frequently purchased products.

The most frequent product in the raw data was:

**WHITE HANGING HEART T-LIGHT HOLDER**

Other frequently occurring products included:

- REGENCY CAKESTAND 3 TIER
- HAND WARMER BABUSHKA DESIGN
- SCOTTIE DOG HOT WATER BOTTLE
- PAPER CHAIN KIT 50'S CHRISTMAS
- HEART OF WICKER SMALL
- CHOCOLATE HOT WATER BOTTLE
- JAM MAKING SET PRINTED
- JUMBO BAG RED RETROSPOT
- HEART OF WICKER LARGE

The same product remained the top-selling/frequently occurring product after cleaning.

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## 10.5 Quantity Analysis

The raw Quantity column had:

- Mean: **8.60**
- Standard deviation: **37.73**

- Minimum: **-2,600**
- Median: **2**
- 75th percentile: **8**
- Maximum: **2,880**

There were **91 non-positive quantity records**.

Negative quantities were treated as invalid (Returns or Cancellations) for the purchase-pattern analysis and removed.

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## 10.6 Price Analysis

The raw Price column had:

- Mean: **4.55**
- Standard deviation: **63.76**
- Minimum: **0**
- Maximum: **13,541.33**

Records with non-positive prices were removed during preprocessing.

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# 11. DATA CLEANING AND PREPROCESSING

Data preprocessing was an important stage because Market Basket Analysis requires clean and consistent transaction data.

The following cleaning operations were performed.

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## 11.1 Duplicate Removal

The project identified **548 duplicate records**.

These were removed using:

```
drop_duplicates()
```

After cleaning:

**Duplicate records = 0**

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## 11.2 Standardization of Text Columns

Whitespace was removed from:

- Itemname
- Country

This ensured consistency in product and country names.

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## 11.3 Handling Missing Item Names

Missing item names were converted to:

**“Unknown”**

This avoided retaining blank product names during subsequent processing.

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## 11.4 Handling Customer IDs

Missing Customer IDs were filled with **0** and converted to an integer-compatible nullable type.

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## 11.5 Date Conversion

Present\_Date was converted from an object/string field into a proper datetime field.

This allows time-based analysis and ensures consistent date handling.

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## 11.6 Removing Invalid Quantities

Records where:

**Quantity  $\leq 0$**

were removed.

A total of:

**91 rows**

were removed because of non-positive quantities.

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## 11.7 Removing Invalid Prices

Records where:

**Price  $\leq 0$**

were removed.

The notebook identified:

**209 records**

with non-positive prices.

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## 12. DATA CLEANING RESULTS

The cleaning process produced the following results:

| Metric         | Raw Dataset                     | Cleaned Dataset      |
|----------------|---------------------------------|----------------------|
| Rows           | 50,000                          | 49,152               |
| Columns        | 7                               | 7                    |
| Missing Values | 18,713                          | 0                    |
| Duplicate Rows | 548                             | 0                    |
| Quantity       | Invalid negative values present | Positive values only |
| Present_Date   | Object                          | Datetime             |
| CustomerID     | Float                           | Integer-compatible   |
| Price          | Included zero/invalid values    | Positive values only |

The EDA report confirms the final cleaned dataset contains **49,152 rows and 7 columns**, with zero missing values and zero duplicates.

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## 13. MARKET BASKET ANALYSIS

Market Basket Analysis is a technique used to identify relationships between products purchased within the same transaction.

The basic idea is:

**If customers frequently purchase Product A and Product B together, the retailer can use that relationship for cross-selling or bundling.**

In this project, each `BillNo` represents a transaction and `Itemname` represents the products purchased within that transaction.

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## 14. TRANSACTION BASKET CREATION

The cleaned dataset was transformed from transaction-level records into a transaction-product matrix.

The project grouped the data by:

- BillNo
- Itemname

and aggregated the corresponding quantity.

This produced a transaction matrix containing:

**1,805 transactions × 2,791 products**

The missing values in this matrix were then replaced with zero.

A binary encoding function was subsequently used:

- 0 = Product not purchased
- 1 = Product purchased

This binary basket format is required for the Apriori algorithm.

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## 15. APRIORI ALGORITHM

The Apriori algorithm is a frequent-itemset mining algorithm used to identify products or groups of products that frequently occur together.

The project used the `mlxtend.frequent_patterns` implementation of Apriori.

The implementation used:

- Minimum support = **0.02**
- Metric for association rules = **Lift**
- Minimum lift threshold = **1**

The code used in the notebook was:

```
frequent_itemset = apriori(basket, min_support=0.02, use_colnames=True)
```

and:

```
rules = association_rules(frequent_itemset, metric="lift", min_threshold=1)
```

---

## 16. ASSOCIATION RULE MINING

Association rules represent relationships between an antecedent and a consequent.

A typical rule can be represented as:

**Product A  $\rightarrow$  Product B**

This means customers who purchase Product A are associated with purchasing Product B.

The project evaluated association rules using:

1. Support
  2. Confidence
  3. Lift
- 

## 17. SUPPORT

Support measures how frequently an item or item combination occurs in the complete transaction dataset.

In simple terms:

**Support = Frequency of the item/itemset across all transactions**

Higher support means that the item combination occurs more frequently.

The notebook describes support as the frequency of an item or combination of items.

The project used:

**Minimum Support = 0.02**

This means itemsets had to meet the selected frequency threshold to be considered frequent.

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## 18. CONFIDENCE

Confidence measures how frequently the consequent occurs when the antecedent is present.

For a rule:

**A  $\rightarrow$  B**

confidence represents the probability of B occurring when A occurs.

The notebook defines:

$$\text{Confidence} = \text{freq}(A,B) / \text{freq}(A)$$

In this project, rules with:

$$\text{Confidence} \geq 0.60$$

were specifically filtered for stronger relationships.

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## 19. LIFT

Lift measures the strength of an association compared with the expected occurrence of the products independently.

Interpretation:

- Lift > 1: Positive association
- Lift = 1: No meaningful association
- Lift < 1: Negative association

The project used:

$$\text{Minimum Lift} = 1$$

The notebook also identifies lift as the metric used to evaluate rule strength.

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## 20. ASSOCIATION RULE RESULTS

The generated rules contained fields such as:

- Antecedents
- Consequents
- Antecedent support
- Consequent support
- Support
- Confidence
- Lift
- Leverage
- Conviction
- Zhang's metric

Some of the strongest observed rules showed very high lift values.

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## 21. RULE FILTERING

To focus on stronger relationships, the project filtered rules using:

- Lift > 1
- Confidence  $\geq 0.60$

This provides a more business-relevant subset of association rules.

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## 22. KEY ANALYTICAL FINDINGS

### 22.1 Customer Buying Behavior

The majority of transactions involve a relatively small number of unique items.

This suggests that many purchases are focused on a limited number of products rather than very large baskets.

The analysis also identified a small group of highly frequent products contributing disproportionately to product demand.

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### 22.2 Best-Selling Product

The most frequently occurring product after cleaning was:

**WHITE HANGING HEART T-LIGHT HOLDER**

The cleaned dataset recorded this product **279 times** in the frequency analysis.

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### 22.3 Product Association Patterns

The Apriori analysis identified product combinations with meaningful associations.

High-support rules represent combinations that occur relatively frequently.

High-confidence rules indicate that the purchase of one product provides a strong indication of purchasing another product.

The project presentation identifies confidence above **0.60** as an important threshold for strong predictive relationships.

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## 22.4 Strong Lift Relationships

Some rules demonstrated exceptionally high lift values.

For example, the notebook contains a rule with:

**Confidence: 0.606557**

and:

**Lift: 21.896721**

Another direction of the same association shows confidence of approximately:

**0.740000**

with the same lift value.

This indicates a much stronger co-occurrence relationship than would be expected if the products were purchased independently.

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## 23. POWER BI DASHBOARD AND VISUALIZATION

Power BI was used to transform analytical outputs into business-friendly visualizations.

The project presentation includes visualizations such as:

### 1. Rules vs Total Lift

This visualization helps identify rules associated with stronger lift values.

### 2. Lift vs Confidence

This visualization allows comparison between:

- Rule strength
- Predictive confidence

### 3. Products Matrix

The product matrix provides a visual representation of product relationships.

The presentation specifically includes these three analytical views.

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## 24. BUSINESS INSIGHTS

The analysis produces several useful business insights.

### **Insight 1: Strong product associations exist**

Certain products are purchased together much more frequently than expected.

This creates opportunities for product recommendations and bundling.

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### **Insight 2: High-confidence rules can support cross-selling**

Rules with confidence greater than 0.60 indicate strong relationships where purchasing one product is frequently associated with purchasing another.

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### **Insight 3: Product bundles can increase transaction value**

The project presentation reports that baskets containing rule-based product pairs showed an estimated **15–30% lift in average order value (AOV)**.

This figure is explicitly described in the presentation as an estimate derived from the association-rule analysis.

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### **Insight 4: A small group of products drives significant demand**

The analysis indicates that a limited group of products appears disproportionately often in transactions.

This supports prioritizing these products in inventory and merchandising decisions.

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### **Insight 5: UK is the dominant market**

The dataset contains 19 countries, but the United Kingdom represents the overwhelming majority of records.

The EDA report of UK as the primary market.

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## **25. BUSINESS RECOMMENDATIONS**

### **25.1 Cross-Selling**

The retailer can use strong association rules to recommend complementary products.

For example:

**Customer purchases Product A → Recommend Product B**

This can be implemented through:

- “Frequently Bought Together”
- Recommended products
- Checkout recommendations
- Product-page suggestions

The project presentation recommends implementing Frequently Bought Together sections based on the strongest association rules.

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## 25.2 Product Bundling

Products with high lift can be combined into promotional bundles.

Example strategy:

**Buy Product X + Product Y together at a discounted price.**

This can encourage customers to purchase multiple products in a single transaction.

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## 25.3 Targeted Marketing

Customers who purchase one product but have not purchased its associated product can be targeted with promotional campaigns.

For example:

**Purchased Product A → Offer discount on Product B**

This approach can improve cross-selling efficiency.

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## 25.4 Inventory Management

Products with high support should receive greater inventory attention because they appear frequently in customer transactions.

Maintaining sufficient stock of these products can reduce stockout risks.

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## 25.5 Reduce Dead Stock

Slow-moving products can potentially be bundled with high-demand products.

This can help improve the movement of less popular products.

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## 26. PROJECT BENEFITS

The project provides several benefits to a retail business.

### Sales Benefits

- Improved cross-selling
- Increased basket size
- Better product bundling
- Improved promotional effectiveness

### Marketing Benefits

- Better product recommendations
- Targeted promotions
- Customer purchase-pattern understanding

### Inventory Benefits

- Better stocking priorities
- Identification of high-demand products
- Potential reduction of slow-moving inventory

### Customer Experience Benefits

- Relevant product recommendations
  - Easier product discovery
  - More useful bundle offers
- 

## 27. CHALLENGES AND LIMITATIONS

### 27.1 Missing Customer Information

A significant number of CustomerID values were missing in the raw dataset.

The project handled these values by assigning 0. This allows the transaction records to remain available for analysis but means those records cannot provide reliable individual-level customer identification.

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### 27.2 Historical Dataset

The dataset covers a relatively short historical period from December 2010 to January 2011.

Therefore, the discovered patterns may not represent current purchasing behavior.

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## 27.3 Product-Level Analysis

The primary focus is product associations.

The project does not perform detailed demographic or behavioral customer segmentation.

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## 27.4 Association Does Not Mean Causation

A high lift or confidence value indicates an association between products.

It does not necessarily mean that purchasing one product causes the purchase of another.

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## 27.5 Computational Complexity

The transaction matrix contains thousands of product columns.

The notebook generated a matrix of approximately **1,805 transactions × 2,791 products**, which can become computationally expensive for frequent-itemset mining.

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# 28. FUTURE SCOPE

The project can be extended in several ways.

## 28.1 Real-Time Recommendation System

The association rules can be integrated into an e-commerce recommendation engine.

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## 28.2 Customer Segmentation

Customer-level segmentation can be performed using:

- RFM Analysis
  - K-Means clustering
  - Customer lifetime value
  - Purchase frequency
-

## 28.3 Advanced Recommendation Models

Future versions could combine Apriori with:

- Collaborative filtering
  - Content-based recommendations
  - Machine learning
  - Deep learning recommendation models
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## 28.4 Time-Based Market Basket Analysis

Product associations could be analyzed separately by:

- Month
- Week
- Day
- Season
- Holiday period

This can identify seasonal purchasing patterns.

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## 28.5 Country-Level Analysis

Association rules could be generated separately for different countries to identify regional purchasing behavior.

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## 28.6 Automated Dashboard

The Power BI dashboard could be connected to regularly updated transaction data so that the business can monitor changing product associations.

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# 29. CONCLUSION

The Purchase Pattern Analysis project successfully demonstrates how retail transaction data can be transformed into meaningful business insights using data analytics and Market Basket Analysis.

The project began with **50,000 raw transaction records** and performed comprehensive data cleaning involving duplicate removal, missing-value handling, date conversion, text standardization, and removal of invalid quantity and price records.

After preprocessing, the dataset contained **49,152 valid records**.

The cleaned transactions were converted into a transaction-product basket and analyzed using the Apriori algorithm.

The project used:

- Minimum support: **0.02**
- Minimum lift threshold: **1**
- Strong-rule confidence filter:  **$\geq 0.60$**

The analysis identified strong relationships between products, including rules with lift values above 20 and confidence values above 0.60.

These insights can support:

- Cross-selling
- Product bundling
- Recommendation systems
- Targeted marketing
- Inventory planning
- Merchandising decisions

Power BI visualization further improves the usability of the analysis by presenting product associations, lift, confidence, and rule relationships in an interactive format.

Overall, the project demonstrates how **Python, Pandas, Market Basket Analysis, the Apriori algorithm, association-rule mining, and Power BI** can be combined to convert raw retail transaction data into actionable business intelligence.

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## 30. REFERENCES

1. Retail Transaction Dataset – 12 PPana.csv
  2. Python Jupyter Notebook – Purchase Pattern Analysis
  3. Cleaned Dataset – df\_clean.xlsx
  4. Exploratory Data Analysis and Data Cleaning Report
  5. Power BI Dashboard – Purchase Pattern Analytics
  6. Mlxtend – Frequent Pattern Mining implementation used for Apriori and association-rule generation.
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# PROJECT SUMMARY

## Project Title

Purchase Pattern Analysis Using Market Basket Analysis and Apriori Algorithm

## Domain

Retail / E-commerce Analytics

## Dataset Size

50,000 records

## Cleaned Dataset

49,152 records

## Number of Columns

7

## Primary Country

United Kingdom

## Top Product

WHITE HANGING HEART T-LIGHT HOLDER

## Main Technique

Market Basket Analysis

## Algorithm

Apriori

## Minimum Support

0.02

## Minimum Lift

1

## Strong Confidence Threshold

0.60

## Programming Language

Python



## Libraries

Pandas, NumPy, Matplotlib, Seaborn, Mlxtend

## Visualization Tool

Power BI

## Key Business Applications

Cross-selling, product bundling, recommendations, targeted marketing, inventory management

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# FINAL PROJECT OUTCOME

The project successfully converts raw retail transaction data into an analytical framework for understanding **what customers buy, which products are purchased together, and how those relationships can be used for business decision-making.**

The strongest value of the project is not simply identifying the best-selling products, but discovering **hidden product relationships through association-rule mining** and translating those relationships into practical retail strategies.